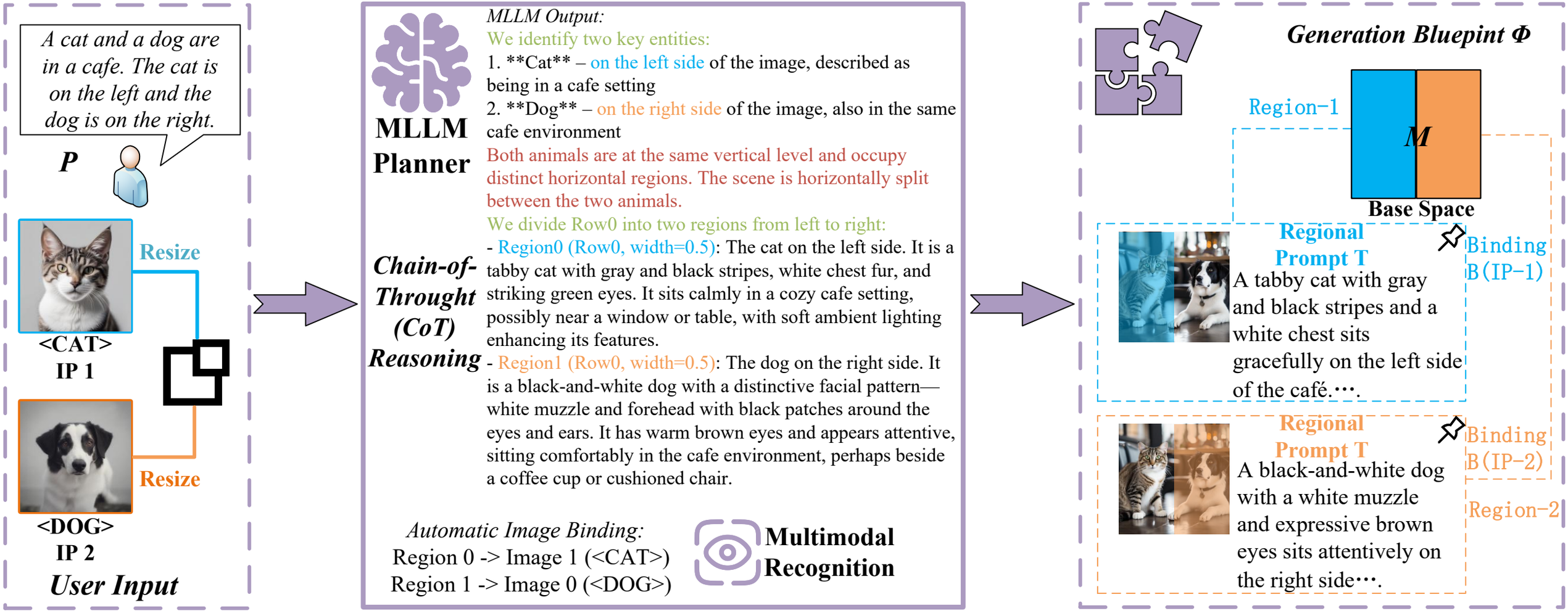
A cat and a dog are in a cafe. The cat is on the left and the dog is on the right.
P
Resize
<CAT>
IP 1
Resize
<DOG>
IP 2
User Input
MLLM Planner
Chain-of-Throught (CoT) Reasoning
MLLM Output:
We identify two key entities:
1. **Cat** – on the left side of the image, described as being in a cafe setting
2. **Dog** – on the right side of the image, also in the same cafe environment
Both animals are at the same vertical level and occupy distinct horizontal regions. The scene is horizontally split between the two animals.
We divide Row0 into two regions from left to right:
- Region0 (Row0, width=0.5): The cat on the left side. It is a tabby cat with gray and black stripes, white chest fur, and striking green eyes. It sits calmly in a cozy cafe setting, possibly near a window or table, with soft ambient lighting enhancing its features.
- Region1 (Row0, width=0.5): The dog on the right side. It is a black-and-white dog with a distinctive facial pattern—white muzzle and forehead with black patches around the eyes and ears. It has warm brown eyes and appears attentive, sitting comfortably in the cafe environment, perhaps beside a coffee cup or cushioned chair.
Automatic Image Binding:
Region 0 -> Image 1 (<CAT>)
Region 1 -> Image 0 (<DOG>)
Multimodal Recognition
Generation Bluepint Φ
Region-1
M
Base Space
Regional Prompt T
A tabby cat with gray and black stripes and a white chest sits gracefully on the left side of the café.⋯.
Binding B(IP-1)
Regional Prompt T
A black-and-white dog with a white muzzle and expressive brown eyes sits attentively on the right side⋯.
Binding B(IP-2)
Region-2